

CHELTENHAM GIRLS HIGH SCHOOL



YEAR 12

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATIONS August 2009

EXTENSION 1 MATHEMATICS

Time allowed : 2 hours (plus 5 minutes reading time).

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work
- Answers are to be completed in blue or black pen on supplied writing paper.
- **Write your name or number** on the top right of each page.
- Board approved calculators may be used.
- Diagrams are not drawn to scale.

Name : _____

Student number : _____

Class Teacher : _____

Question	1	2	3	4	5	6	7	Total	%
Mark	/12	/12	/12	/12	/12	/12	/12	/84	

Question 1	Start a new page.	Marks
a)	Differentiate $y = \cos^{-1} 3x$	2
b)	Find $\int \cos^2 x \, dx$	2
c)	Show that $F(x) = \frac{\sqrt{1-x^2}}{\sin^{-1} x}$ is an odd function.	2
d)	Find the coordinates of the point P which divides the interval AB with end points A(2, 3) and B(7, -7) externally in the ratio 4:9.	3
e)	Evaluate $\int_1^{\sqrt{3}} \frac{x}{x^2+1} \, dx$, giving your answers in simplified, exact form.	3

Question 2	Start a new page.	Marks
a)	i) Factorise $a^3 - b^3$.	1
	ii) Hence or otherwise show that if $\sin A \neq \cos A$ then $\frac{2 \sin^3 A - 2 \cos^3 A}{\sin A - \cos A} = 2 + \sin 2A$.	3
b)	Solve the inequality $\frac{2x-3}{x+2} \geq 3$	3
c)	Use calculus to find the value of x which will maximise the value of $3 \sin x + 2 \cos x$. Give your answer in radians, correct to 2 decimal places. (You need not find the maximum value or prove that it will actually be a maximum.)	2
d)	Use $x = \log u$ to find $\int \frac{e^x}{\sqrt{1-e^{2x}}} \, dx$.	3

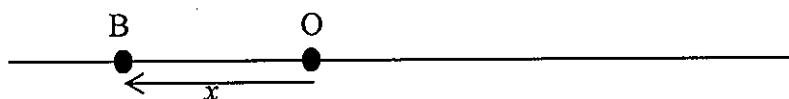
Question 3 is on the next page.

Question 3**Start a new page.****Marks**

- a) A polynomial is given by $f(x) = x^3 + ax^2 + 7x + b$. Find the values of a and b if $(x+3)$ is a factor of $f(x)$ and when $f(x)$ is divided by $(x-2)$ the remainder is 35. 4
- b) The equation $\ln x + x = 2$ has only one root.
 i) Show that the root lies between $x = 1$ and $x = 2$. 1
 ii) With 2 as a first estimate of the root, use one application of Newton's method to find an other approximation to the root, correct to 2 decimal places. 3
- c) Use the method of mathematical induction to prove that $9^{n+2} - 4^n$ is divisible by 5 for all positive integers n . 4

Question 4**Start a new page.****Marks**

- a) Let $f(x) = 2x - x^2$ for $x \leq 1$. This function has an inverse $f^{-1}(x)$.
 i) Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the same diagram. 2
 ii) Find an expression for $f^{-1}(x)$. 3
 iii) Evaluate $f^{-1}\left(\frac{3}{4}\right)$. 2
- b) The point B is allowed to move along a fixed line such that its acceleration in cm/s^2 is given by $\ddot{x} = x^2 + 3$ where x is the displacement of OB.



Initially B is at rest, 3cm to the left of O (where $x = -3$).

- i) Prove that B will move with velocity v such that $v^2 = \frac{2}{3}x^3 + 6x + 36$. 2
- ii) By the time that B is 3cm to the right of O, how fast is it moving, to the nearest 0.01cm/s? 2
- iii) Explain how we know that B will never move towards the left. 1

Question 5**Start a new page.****Marks**

- a) Mr and Mrs Bunn, Mr and Mrs Dunn, Mr and Mrs Funn, Mr and Mrs Gunn and Mr and Mrs Hunn are a group of 10 friends at a fund-raising dance in aid of the bush fire victims.
How many ways can :
- i) they stand in line to do the Macarena? (There are no restrictions on who stands beside whom.) 1
 - ii) they sit around a round table if no men sit together? 2
 - iii) 9 of them be chosen to dance a particular dance which requires exactly 9 people? 1
- b) A particle is moving in simple harmonic motion on a line. Its maximum acceleration is $2m/s^2$ and its maximum speed is $6 m/s$. Find the amplitude and period of its motion. 2
- c) When a stone is projected from the origin with a velocity V , directed α degrees above horizontal, the equations of motion horizontally and vertically are given by
$$x = Vt \cos \alpha \text{ and } y = Vt \sin \alpha - 5t^2.$$
 Do NOT prove these results.
- i) Show that the maximum height of its flight path is $\frac{V^2 \sin^2 \alpha}{20}$. 2
 - ii) Prove that the stone will land furthest from the origin when $\alpha = 45^\circ$. 3
 - iii) In terms of V , what is this maximum range of the stone? 1

Question 6**Start a new page.****Marks**

- a) Sketch $y = \frac{x}{x^2 - 4}$, clearly indicating its asymptotes. 4
- b) The line $y = x + b$ intersects $x^2 = 12y$ at $P(x_1, y_1)$ and $Q(x_2, y_2)$.
 M is the midpoint of interval PQ .
- i) Show this information on a neat sketch. 1
 - ii) Find the equation of the locus of M . 5
 - iii) Find the values of b for which P and Q are 2 distinct points. 2

Question 7

Start a new page.

Marks

a) i) Use the result for $\cos(A + B)$ to prove that $\cos 2x = 1 - 2\sin^2 x$.

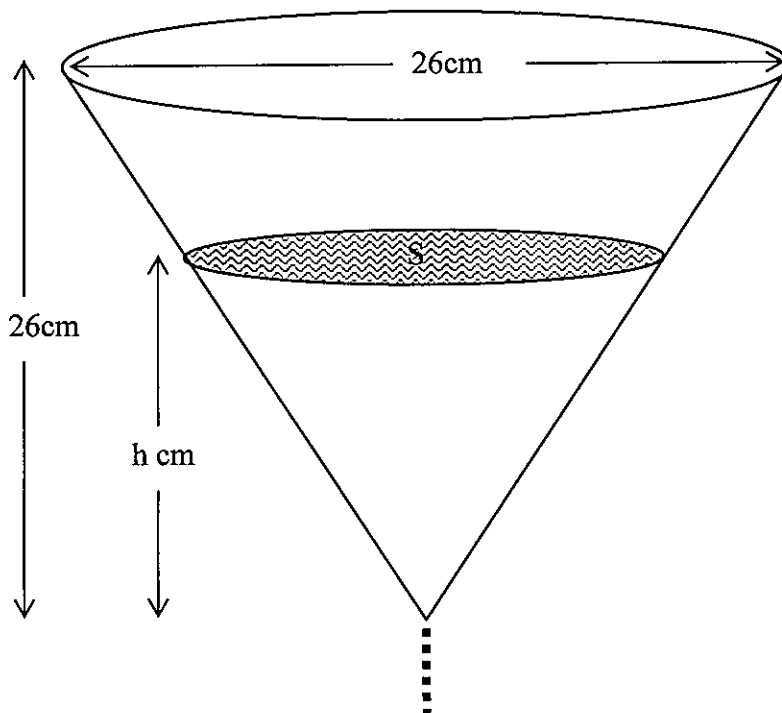
1

ii) Use the result $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ to find $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$.

2

b) A conical vessel has height 26cm and diameter 26cm .

Water is leaking out of the bottom at a rate of 80cm^3 per minute. After t minutes the depth of water is $h\text{cm}$.



i) Find the surface area S (shaded) of the water in terms of h .

1

ii) Show that the volume of water is given by

1

$$V = \frac{1}{12}\pi h^3.$$

iii) Find expressions for $\frac{dV}{dh}$, $\frac{dS}{dh}$ and $\frac{dV}{dt}$.

3

iv) Find the rate at which the surface area S is decreasing when the water is 10cm deep.

4

End of test.

$$\textcircled{1} \text{ a) } y' = \frac{-1}{\sqrt{1-9x^2}} \cdot 3$$

$$= \frac{-3}{\sqrt{1-9x^2}}$$

$$\text{b) } \int \cos^2 x \, dx = \frac{1}{2} \int (1 + \cos 2x) \, dx$$

$$= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right] + C$$

$$\text{c) } F(-x) = \frac{\sqrt{1-(-x)^2}}{\sin^{-1}(-x)}$$

$$= \frac{\sqrt{1-x^2}}{-\sin^{-1} x}$$

$$= -F(x)$$

$\therefore F(x)$ is odd.

$$\text{d) } A(2, 3) \quad -4:9 \quad B(7, -7)$$

$$P = \left(\frac{9 \times 2 - 4 \times 7}{9 - 4}, \frac{9 \times 3 - 4 \times -7}{9 - 4} \right)$$

$$= \left(\frac{18 - 28}{5}, \frac{27 + 28}{5} \right)$$

$$= (-2, 11)$$

$$\text{e) } \frac{1}{2} \int_1^{\sqrt{5}} \frac{2x}{x^2 + 1} \, dx = \frac{1}{2} \left[\ln(x^2 + 1) \right]_1^{\sqrt{5}}$$

$$= \frac{1}{2} \ln 4 - \frac{1}{2} \ln 2$$

$$= \frac{1}{2} \ln 2$$

$$(2) a) i) a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$ii) \frac{2(\sin^3 A - \cos^3 A)}{\sin A - \cos A} = \frac{2(\sin A - \cos A)(\sin^2 A + \sin A \cos A + \cos^2 A)}{\sin A - \cos A}$$

$$= 2(1 + \sin A \cos A)$$

$$= 2 + \sin 2A$$

$$b) \therefore (2x-3)(x+2) \geq 3(x+2)^2$$

$$2x^2 + x - 6 \geq 3x^2 + 12x + 12$$

$$x^2 + 11x + 18 \leq 0$$

$$(x+9)(x+2) \leq 0$$

$$-9 \leq x \leq -2$$

But denominator $\neq 0$

$$\therefore x \neq -2$$

$$\therefore -9 \leq x < -2$$

$$c) y = 3 \sin x + 2 \cos x$$

$$\therefore y' = 3 \cos x - 2 \sin x$$

For max, $y' = 0$

$$\therefore 3 \cos x = 2 \sin x$$

$$\therefore \frac{3}{2} = \frac{\sin x}{\cos x}$$

$$= \tan x$$

$$\therefore x = 0.98279 \dots$$

$$\approx 0.98$$

$$d) x = \log u \Rightarrow u = e^x$$

$$\therefore dx = \frac{1}{u} du$$

$$\therefore \int \frac{e^x}{\sqrt{1-e^{2x}}} dx = \int \frac{u}{\sqrt{1-u^2}} \cdot \frac{1}{u} du$$

$$= \sin^{-1} u + C$$

$$= \sin^{-1} e^x + C$$

$$\textcircled{3} \text{ a) } f(x) = x^3 + ax^2 + 7x + b$$

$$f(-3) = 0$$

$$\therefore 0 = -27 + 9a - 21 + b$$

$$0 = 9a + b - 48 \dots \textcircled{1}$$

$$f(2) = 35$$

$$\therefore 35 = 8 + 4a + 14 + b$$

$$0 = 4a + b - 13 \dots \textcircled{2}$$

$$\textcircled{1} - \textcircled{2}: 0 = 5a - 35$$

$$a = 7$$

$$\textcircled{1}: \therefore 0 = 63 + b - 48$$

$$b = -15$$

$$a = 7$$

$$\text{b) i) Let } \ln x + x - 2 = f(x)$$

$$f(1) = \ln 1 + 1 - 2$$

$$= -1$$

$$< 0$$

$$f(2) = \ln 2 + 2 - 2$$

$$= \ln 2$$

$$> 0$$

$\therefore$ There is a root between 1 & 2.

$$\text{ii) } f'(x) = \frac{1}{x} + 1$$

$$\text{If } x_1 = 2, \quad x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 2 - \frac{\ln 2 + 2 - 2}{\frac{1}{2} + 1}$$

$$= 2 - \frac{2 \ln 2}{3}$$

$$= 1.53790188$$

$$\approx 1.54$$

③ c) RTP: $9^{n+2} - 4^n = 5A_i$ (A_i integral)

Try $n=1$: $LHS = 9^3 - 4^1$
 $= 725$
 $= 5 \times 145$

$\therefore$ True for $n=1$

Assume true for $n=k$ (k integral)

i.e. $9^{k+2} - 4^k = 5A_k \dots \textcircled{1}$

Try for $n=k+1$:

$$\begin{aligned} 9^{k+3} - 4^{k+1} &= 9 \times 9^{k+2} - 4 \times 4^k \\ &= 9(5A_k + 4^k) - 4 \times 4^k \\ &= 45A_k + 9 \times 4^k - 4 \times 4^k \\ &= 5(9A_k + 4^k) \end{aligned}$$

$\therefore$ If true for $n=k$ then true for $n=k+1$

But true for $n=1$

$\therefore$ " " $n=2$

$\therefore$ " " $n=3$

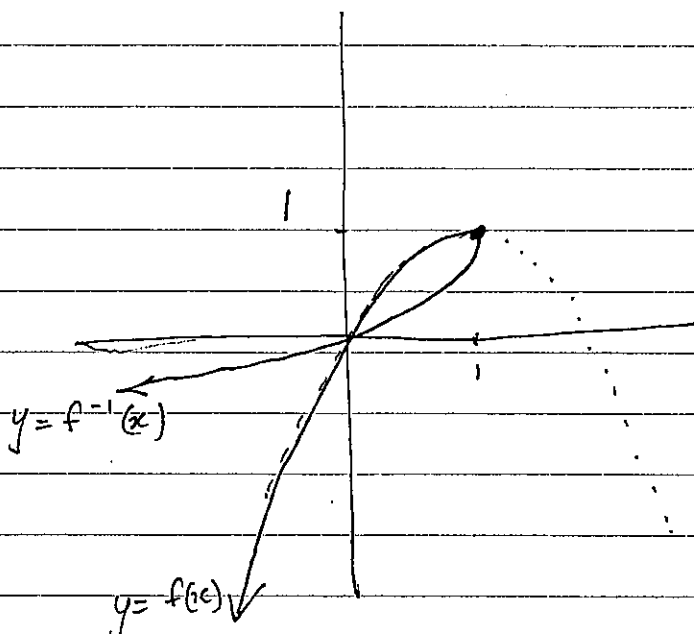
etc.

$\therefore$ True for all positive integers.

4 a) i)

$$f(x) = x(2-x)$$

$$f(1) = 2 - 1 = 1$$



ii) $x = 2y - y^2$

$$\therefore y^2 - 2y + x = 0$$

$$y = \frac{2 \pm \sqrt{4 - 4x}}{2 \times 1}$$

$$= 1 \pm \sqrt{1-x}$$

Inverse is $f^{-1}(x) = 1 - \sqrt{1-x}$

iii) $f^{-1}\left(\frac{3}{4}\right) = 1 - \sqrt{1 - \frac{3}{4}}$

$$= 1 - \frac{1}{2}$$

$$= \frac{1}{2}$$

b) $\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right) = x^2 + 3$

$$\therefore \frac{1}{2}v^2 = \frac{1}{3}x^3 + 3x + c$$

When $x = -3$, $v = 0$

$$\therefore 0 = -9 - 9 + c$$

$$c = 18$$

$$\therefore v^2 = \frac{2}{3}x^3 + 6x + 36$$

Q.E.D.

ii) When $x = 3$, $v^2 = \frac{2}{3}(3)^3 + 6(3) + 36$
 $= 72$

$$\therefore v = 8.48528137...$$

$$\approx 8.49 \text{ cm/s.}$$

iii) $\ddot{x} = x^2 + 3$ is always positive and B is initially at rest. $\therefore$ B will move to the right, getting faster always and thus will never move left.

iii)

⑤ a) ~~n~~ (one left out) = 10

i) ~~n~~ (10 in line) = 10!

(= 3 628 800)

ii) ~~n~~ (alternate around table)

= n (men around table) × n (women between them)

= 4! × 5!

(= 2880)

[Award one mark for using 9!, 4! or 5! in working.]

[or one mark for indicating that they need to alternate.]

b) Let $x = A \sin nt$ for SHM

$\therefore \dot{x} = An \cos nt$, which has maximum of An

$\therefore \ddot{x} = -An^2 \sin nt$, " " " " An^2

$\therefore An^2 = 2 \dots \textcircled{1}$

$An = 6 \dots \textcircled{2}$

$n = \frac{1}{3}$

$\textcircled{2}: \frac{1}{3}A = 6$

$\therefore A = 18$

$T = \frac{2\pi}{n}$

$= \frac{2\pi}{\frac{1}{3}}$

$= 6\pi$

$\therefore$ Amplitude = 18 cm, period = 6π seconds.

c) i) $y = Vt \sin \alpha - 5t^2$

$\therefore \dot{y} = V \sin \alpha - 10t$

For max y , $\dot{y} = 0 \therefore V \sin \alpha = 10t$

$t = \frac{V \sin \alpha}{10}$

$\therefore \text{Max } y = V \left(\frac{V \sin \alpha}{10} \right) \sin \alpha - 5 \left(\frac{V \sin \alpha}{10} \right)^2$

$= \frac{V^2 \sin^2 \alpha}{10} - \frac{5 V^2 \sin^2 \alpha}{100}$

$= \frac{V^2 \sin^2 \alpha}{20}$

QED.

⑤ c ii) When $y=0$, $Vt \sin \alpha - 5t^2 = 0$
 $t(V \sin \alpha - 5t) = 0$

$\therefore t=0$ or $t = \frac{V \sin \alpha}{5}$

When $t = \frac{V \sin \alpha}{5}$, $x = \frac{V^2 \sin \alpha \cos \alpha}{5}$

$= \frac{V^2 \sin 2\alpha}{10} \dots \textcircled{1}$

which has maximum when $\sin 2\alpha = 1$

$\therefore 2\alpha = 90^\circ$

$\therefore \alpha = 45^\circ$

R.E.D.

iii) ~~Subst. $\alpha = 45^\circ$ into range: $x = V$~~

From (ii), maximum range $= \frac{V^2}{10}$.

⑥ a) For vertical asymptotes, $x^2 - 4 = 0$

$$\therefore x = \pm 2$$

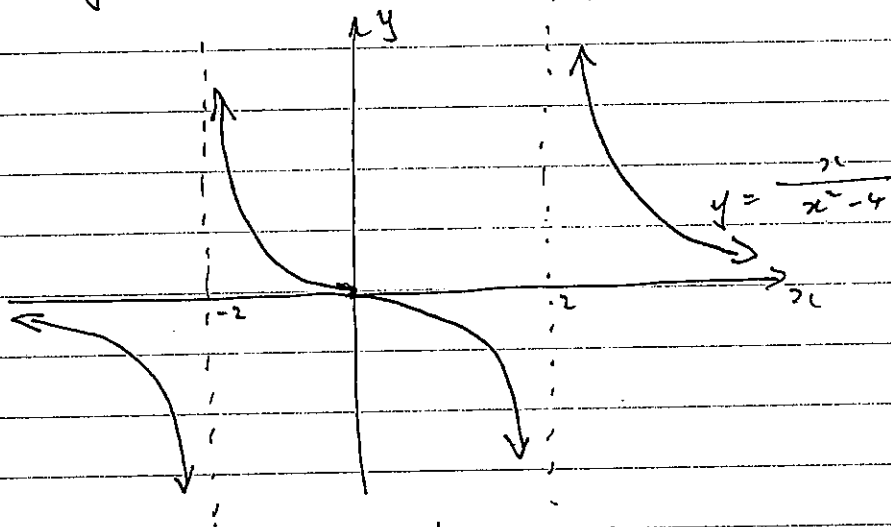
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$$\lim_{x \rightarrow \pm \infty} \frac{x}{x^2 - 4} = \lim_{x \rightarrow \pm \infty} \frac{\frac{1}{x}}{1 - \frac{4}{x^2}}$$

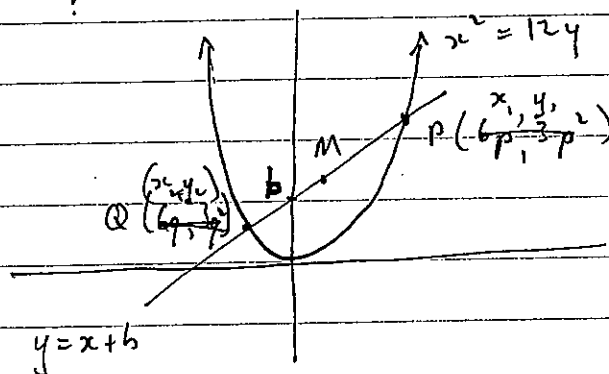
$$= \frac{0}{1 - 0}$$

$$= 0$$

$\therefore y = 0$ is horizontal asymptote. #



b) i)



$$ii) M = \left(\frac{6p + 6q}{2}, \frac{3p^2 + 3q^2}{2} \right)$$

$$x^2 = 12y$$

$$y = x + b$$

$$\therefore x^2 = 12x + 12b$$

①

$$x^2 - 12x - 12b = 0$$

$$\therefore x = \frac{12 \pm \sqrt{144 + 48b}}{2}$$

$$= 6 \pm \sqrt{36 + 12b} \equiv x_1, x_2$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{6 + \sqrt{\dots} + 6 - \sqrt{\dots}}{2}, \frac{6 + \sqrt{\dots} + b + 6 - \sqrt{\dots} + b}{2} \right)$$

$$= (6, 6 + b)$$

$\therefore$ Locus of M is $x = 6$

iii) For distinct points, in (i), $\Delta > 0$

$$\therefore 144 + 48b > 0$$

$$48b > -144$$

$$b > -3$$

$$(7) a) i) \cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\therefore \cos 2x = \cos x \cos x - \sin x \sin x$$

$$= \cos^2 x - \sin^2 x$$

$$= 1 - 2\sin^2 x$$

$$(\because \sin^2 x + \cos^2 x = 1)$$

$$ii) \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = \lim_{x \rightarrow 0} \frac{1 - (1 - 2\sin^2 x)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{2\sin^2 x}{x^2}$$

$$= 2 \times 1^2$$

$$= 2$$

$$b) i) \text{Diam.} = h$$

$$\therefore \text{rad.} = \frac{h}{2}$$

$$\begin{aligned} \therefore S &= \pi r^2 \\ &= \pi \left(\frac{h}{2}\right)^2 \\ &= \frac{\pi h^2}{4} \end{aligned}$$

$$ii) V = \frac{1}{3} S h$$

$$= \frac{1}{3} \times \frac{\pi h^2}{4} \times h$$

$$= \frac{1}{12} \pi h^3 \quad \text{QED}$$

$$iii) \frac{dV}{dh} = \frac{1}{4} \pi h^2$$

$$\frac{dS}{dh} = \frac{1}{2} \pi h$$

$$\frac{dV}{dt} = -80$$

[accept +80]

$$iv) \frac{dS}{dt} = \frac{dS}{dh} \cdot \frac{dh}{dV} \cdot \frac{dV}{dt}$$

$$= \frac{\frac{1}{2} \pi h}{2} \cdot \frac{4}{\pi h^2} \cdot \frac{-80}{1}$$

$$= \frac{-160 \pi h}{\pi h^2}$$

$$\begin{aligned} \text{When } h &= 10, \quad \frac{dS}{dt} = \frac{-160}{10} \\ &= -16 \end{aligned}$$

ie. surface decreasing by $16 \text{ cm}^2/\text{min}$.